



Lab Report – EE-170L Computer Programming (Lab 9)

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Objectives

- Understand the concept of functions in C++.
- Learn how to declare, define, and call functions.
- Practice writing functions with and without parameters.
- Apply functions to solve problems using arrays.

Theory

Functions in C++ allow us to break programs into smaller reusable parts. Each function has:

- A **name** (identifier).
- **Parameters** (optional inputs).
- A **return type** (the kind of value it gives back).

The **prototype** tells the compiler what the function looks like before it is used.

Syntax:

`returnType functionName(parameters);` **Examples:**

- `int main();` → returns an integer, no parameters.
- `float divide(int a, int b);` → returns a float, takes two integers.
- `void square(float x);` → returns nothing, takes one float.

Lab Tasks

Task 1: Function Introduction and Declaration

```
#include <iostream> // include input/output library using
namespace std; // use standard namespace void greet() {//
function declaration with no parameters

    cout << "Welcome to Lab 09!"; // print greeting message
}
```

```
int main() { // main function starts greet(); //  
    call the greet function return 0; // end  
    program  
}
```

Task 2: Basic Function Declaration and Call

```
#include <iostream> // include input/output library using  
namespace std; // use standard namespace
```

```
void sayHello() { // function declaration with no parameters  
    cout << "Hello, World!"; // print hello message  
}
```

```
int main() { // main function starts  
    sayHello(); // call the sayHello function return 0; //  
    end program  
}
```

Task 3: Sum and Average of Array Elements

```
#include <iostream> // include input/output library using  
namespace std; // use standard namespace
```

```
int calculateSum(int arr[], int size) { // function to calculate sum int  
    sum = 0; // variable to store sum
```



```

for(int i = 0; i < size; i++) {           // loop through array
    sum += arr[i];                       // add each element to sum
}

return sum;                             // return total sum
}

int main() {                             // main function starts
int numbers[10];                         // declare array of size 10
    cout << "Enter 10 integers: "; // ask user for input
    for(int i = 0; i < 10; i++) { // loop for 10 inputs
        cin >> numbers[i];         // store each input in array
    }

int total = calculateSum(numbers, 10); // call function to get sum
float average = total / 10.0; // calculate average

cout << "Sum = " << total << endl; // display sum
cout << "Average = " << average << endl; // display average
return 0;

// end program
}

```

Results

- Task 1 displayed: Welcome to Lab 09!

- Task 2 displayed: Hello, World!
- Task 3 correctly calculated the sum and average of 10 integers entered by

the user. **Conclusion**

In this lab, I learned how to:

- Declare and define functions in C++.
- Call functions from main().
- Pass parameters to functions and return values.
- Use functions to simplify array operations like sum and average.

Functions make programs **modular, reusable, and easier to understand**.